

LU Decomposition Method

System of Linear Equations:

$$\mathbf{Ax} = \mathbf{b} \rightarrow \mathbf{LUx} = \mathbf{b} \rightarrow \mathbf{A} = \mathbf{LU}$$

L is lower triangular matrix.

U is upper triangular matrix.

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & \cdots & 0 \\ l_{21} & l_{22} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ l_{n1} & \cdots & l_{n,n-1} & l_{nn} \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & \cdots & u_{1n} \\ 0 & u_{22} & \ddots & \vdots \\ \vdots & \ddots & \ddots & u_{n-1,n} \\ 0 & \cdots & 0 & u_{nn} \end{bmatrix}$$

$$\mathbf{Ly} = \mathbf{b} \rightarrow \begin{bmatrix} l_{11} & 0 & \cdots & 0 \\ l_{21} & l_{22} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ l_{n1} & \cdots & l_{n,n-1} & l_{nn} \end{bmatrix} \begin{Bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{Bmatrix} \rightarrow \begin{aligned} y_1 &= \frac{b_1}{l_{11}} \\ y_2 &= \frac{b_2 - l_{21} \cdot y_1}{l_{22}} \\ &\vdots \end{aligned}$$

$$\mathbf{Ux} = \mathbf{y} \rightarrow \begin{bmatrix} u_{11} & u_{12} & \cdots & u_{1n} \\ 0 & u_{22} & \ddots & \vdots \\ \vdots & \ddots & \ddots & u_{n-1,n} \\ 0 & \cdots & 0 & u_{nn} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{Bmatrix} \rightarrow \begin{aligned} x_n &= \frac{y_n}{u_{nn}} \\ x_{n-1} &= \frac{y_{n-1} - u_{n-1,n} \cdot x_n}{l_{22}} \\ &\vdots \end{aligned}$$

LU Decomposition Method

Crout Matrix Decomposition:

LU Decomposition Method

Crout Matrix Decomposition:

$$\mathbf{L} = \begin{bmatrix} l_{11} & 0 & \cdots & 0 \\ l_{21} & l_{22} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ l_{n1} & \cdots & l_{n,n-1} & l_{nn} \end{bmatrix}, \mathbf{U} = \begin{bmatrix} \color{red}{1} & u_{12} & \cdots & u_{1n} \\ 0 & \color{red}{1} & \ddots & \vdots \\ \vdots & \ddots & \color{red}{\ddots} & u_{n-1,n} \\ 0 & \cdots & 0 & \color{red}{1} \end{bmatrix}$$

LU Decomposition Method

Crout Matrix Decomposition:

$$\mathbf{L} = \begin{bmatrix} l_{11} & 0 & \cdots & 0 \\ l_{21} & l_{22} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ l_{n1} & \cdots & l_{n,n-1} & l_{nn} \end{bmatrix}, \mathbf{U} = \begin{bmatrix} \color{red}{1} & u_{12} & \cdots & u_{1n} \\ 0 & \color{red}{1} & \ddots & \vdots \\ \vdots & \ddots & \color{red}{\ddots} & u_{n-1,n} \\ 0 & \cdots & 0 & \color{red}{1} \end{bmatrix}$$

Finding Elements of $\mathbf{L}$ and $\mathbf{U}$:

LU Decomposition Method

Crout Matrix Decomposition:

$$\mathbf{L} = \begin{bmatrix} l_{11} & 0 & \cdots & 0 \\ l_{21} & l_{22} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ l_{n1} & \cdots & l_{n,n-1} & l_{nn} \end{bmatrix}, \mathbf{U} = \begin{bmatrix} \color{red}{1} & u_{12} & \cdots & u_{1n} \\ 0 & \color{red}{1} & \ddots & \vdots \\ \vdots & \ddots & \ddots & u_{n-1,n} \\ 0 & \cdots & 0 & \color{red}{1} \end{bmatrix}$$

Finding Elements of $\mathbf{L}$ and $\mathbf{U}$:

$$\mathbf{A} = \mathbf{LU}$$

$$\begin{bmatrix} l_{11} & 0 & 0 & 0 \\ l_{21} & l_{22} & 0 & 0 \\ l_{31} & l_{32} & l_{33} & 0 \\ l_{41} & l_{42} & l_{43} & l_{44} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} & u_{14} \\ 0 & 1 & u_{23} & u_{24} \\ 0 & 0 & 1 & u_{34} \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

LU Decomposition Method

Crout Matrix Decomposition:

$$\mathbf{L} = \begin{bmatrix} l_{11} & 0 & \cdots & 0 \\ l_{21} & l_{22} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ l_{n1} & \cdots & l_{n,n-1} & l_{nn} \end{bmatrix}, \mathbf{U} = \begin{bmatrix} \mathbf{1} & u_{12} & \cdots & u_{1n} \\ 0 & \mathbf{1} & \ddots & \vdots \\ \vdots & \ddots & \ddots & u_{n-1,n} \\ 0 & \cdots & 0 & \mathbf{1} \end{bmatrix}$$

Finding Elements of $\mathbf{L}$ and $\mathbf{U}$:

$$\mathbf{A} = \mathbf{LU}$$

$$\begin{bmatrix} \mathbf{l}_{11} & 0 & 0 & 0 \\ l_{21} & l_{22} & 0 & 0 \\ l_{31} & l_{32} & l_{33} & 0 \\ l_{41} & l_{42} & l_{43} & l_{44} \end{bmatrix} \begin{bmatrix} \mathbf{1} & u_{12} & u_{13} & u_{14} \\ 0 & 1 & u_{23} & u_{24} \\ 0 & 0 & 1 & u_{34} \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

$$l_{11} \cdot 1 + 0 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 = a_{11}$$

$$\mathbf{l}_{11} = \mathbf{a}_{11}$$

LU Decomposition Method

Crout Matrix Decomposition:

$$\mathbf{L} = \begin{bmatrix} l_{11} & 0 & \cdots & 0 \\ l_{21} & l_{22} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ l_{n1} & \cdots & l_{n,n-1} & l_{nn} \end{bmatrix}, \mathbf{U} = \begin{bmatrix} \mathbf{1} & u_{12} & \cdots & u_{1n} \\ 0 & \mathbf{1} & \ddots & \vdots \\ \vdots & \ddots & \ddots & u_{n-1,n} \\ 0 & \cdots & 0 & \mathbf{1} \end{bmatrix}$$

Finding Elements of $\mathbf{L}$ and $\mathbf{U}$:

$$\mathbf{A} = \mathbf{LU}$$

$$\begin{bmatrix} l_{11} & 0 & 0 & 0 \\ \mathbf{l_{21}} & \mathbf{l_{22}} & \mathbf{0} & \mathbf{0} \\ l_{31} & l_{32} & l_{33} & 0 \\ l_{41} & l_{42} & l_{43} & l_{44} \end{bmatrix} \begin{bmatrix} \mathbf{1} & u_{12} & u_{13} & u_{14} \\ \mathbf{0} & 1 & u_{23} & u_{24} \\ \mathbf{0} & 0 & 1 & u_{34} \\ \mathbf{0} & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ \mathbf{a_{21}} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

$$l_{21} \cdot 1 + l_{22} \cdot 0 + 0 \cdot 0 + 0 \cdot 0 = a_{21}$$

$$\mathbf{l_{21} = a_{21}}$$

LU Decomposition Method

Crout Matrix Decomposition:

$$\mathbf{L} = \begin{bmatrix} l_{11} & 0 & \cdots & 0 \\ l_{21} & l_{22} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ l_{n1} & \cdots & l_{n,n-1} & l_{nn} \end{bmatrix}, \mathbf{U} = \begin{bmatrix} \mathbf{1} & u_{12} & \cdots & u_{1n} \\ 0 & \mathbf{1} & \ddots & \vdots \\ \vdots & \ddots & \ddots & u_{n-1,n} \\ 0 & \cdots & 0 & \mathbf{1} \end{bmatrix}$$

Finding Elements of $\mathbf{L}$ and $\mathbf{U}$:

$$\mathbf{A} = \mathbf{LU}$$

$$\begin{bmatrix} l_{11} & 0 & 0 & 0 \\ l_{21} & l_{22} & 0 & 0 \\ \mathbf{l_{31}} & \mathbf{l_{32}} & \mathbf{l_{33}} & \mathbf{0} \\ l_{41} & l_{42} & l_{43} & l_{44} \end{bmatrix} \begin{bmatrix} \mathbf{1} & u_{12} & u_{13} & u_{14} \\ \mathbf{0} & 1 & u_{23} & u_{24} \\ \mathbf{0} & 0 & 1 & u_{34} \\ \mathbf{0} & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ \mathbf{a_{31}} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

$$l_{31} \cdot 1 + l_{32} \cdot 0 + l_{33} \cdot 0 + 0 \cdot 0 = a_{31}$$

$$\mathbf{l_{31} = a_{31}}$$

LU Decomposition Method

Crout Matrix Decomposition:

$$\mathbf{L} = \begin{bmatrix} l_{11} & 0 & \cdots & 0 \\ l_{21} & l_{22} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ l_{n1} & \cdots & l_{n,n-1} & l_{nn} \end{bmatrix}, \mathbf{U} = \begin{bmatrix} \mathbf{1} & u_{12} & \cdots & u_{1n} \\ 0 & \mathbf{1} & \ddots & \vdots \\ \vdots & \ddots & \ddots & u_{n-1,n} \\ 0 & \cdots & 0 & \mathbf{1} \end{bmatrix}$$

Finding Elements of $\mathbf{L}$ and $\mathbf{U}$:

$$\mathbf{A} = \mathbf{LU}$$

$$\begin{bmatrix} l_{11} & 0 & 0 & 0 \\ l_{21} & l_{22} & 0 & 0 \\ l_{31} & l_{32} & l_{33} & 0 \\ \mathbf{l_{41}} & \mathbf{l_{42}} & \mathbf{l_{43}} & \mathbf{l_{44}} \end{bmatrix} \begin{bmatrix} \mathbf{1} & u_{12} & u_{13} & u_{14} \\ \mathbf{0} & 1 & u_{23} & u_{24} \\ \mathbf{0} & 0 & 1 & u_{34} \\ \mathbf{0} & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ \mathbf{a_{41}} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

$$l_{41} \cdot 1 + l_{42} \cdot 0 + l_{43} \cdot 0 + l_{44} \cdot 0 = a_{41}$$

$$\mathbf{l_{41} = a_{41}}$$

LU Decomposition Method

Crout Matrix Decomposition:

$$\mathbf{L} = \begin{bmatrix} l_{11} & 0 & \cdots & 0 \\ l_{21} & l_{22} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ l_{n1} & \cdots & l_{n,n-1} & l_{nn} \end{bmatrix}, \mathbf{U} = \begin{bmatrix} \mathbf{1} & u_{12} & \cdots & u_{1n} \\ 0 & \mathbf{1} & \ddots & \vdots \\ \vdots & \ddots & \ddots & u_{n-1,n} \\ 0 & \cdots & 0 & \mathbf{1} \end{bmatrix}$$

Finding Elements of $\mathbf{L}$ and $\mathbf{U}$:

$$\mathbf{A} = \mathbf{LU}$$

$$\begin{bmatrix} \mathbf{l}_{11} & 0 & 0 & 0 \\ l_{21} & l_{22} & 0 & 0 \\ l_{31} & l_{32} & l_{33} & 0 \\ l_{41} & l_{42} & l_{43} & l_{44} \end{bmatrix} \begin{bmatrix} 1 & \mathbf{u}_{12} & u_{13} & u_{14} \\ 0 & \mathbf{1} & u_{23} & u_{24} \\ 0 & 0 & 1 & u_{34} \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & \mathbf{a}_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

$$l_{11} \cdot u_{12} + 0 \cdot 1 + 0 \cdot 0 + 0 \cdot 0 = a_{12}$$

$$\mathbf{u}_{12} = \mathbf{a}_{12} / \mathbf{l}_{11}$$

LU Decomposition Method

Crout Matrix Decomposition:

$$\mathbf{L} = \begin{bmatrix} l_{11} & 0 & \cdots & 0 \\ l_{21} & l_{22} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ l_{n1} & \cdots & l_{n,n-1} & l_{nn} \end{bmatrix}, \mathbf{U} = \begin{bmatrix} \color{red}{1} & u_{12} & \cdots & u_{1n} \\ 0 & \color{red}{1} & \ddots & \vdots \\ \vdots & \ddots & \ddots & u_{n-1,n} \\ 0 & \cdots & 0 & \color{red}{1} \end{bmatrix}$$

Finding Elements of $\mathbf{L}$ and $\mathbf{U}$:

$$\mathbf{A} = \mathbf{LU}$$

$$\begin{bmatrix} \color{red}{l_{11}} & \color{red}{0} & \color{red}{0} & \color{red}{0} \\ l_{21} & l_{22} & 0 & 0 \\ l_{31} & l_{32} & l_{33} & 0 \\ l_{41} & l_{42} & l_{43} & l_{44} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & \color{red}{u_{13}} & u_{14} \\ 0 & 1 & \color{red}{u_{23}} & u_{24} \\ 0 & 0 & \color{red}{1} & u_{34} \\ 0 & 0 & \color{red}{0} & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & \color{red}{a_{13}} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

$$l_{11} \cdot u_{13} + 0 \cdot u_{23} + 0 \cdot 1 + 0 \cdot 0 = a_{13}$$

$$\color{red}{u_{13}} = \color{red}{a_{13} / l_{11}}$$

LU Decomposition Method

Crout Matrix Decomposition:

$$\mathbf{L} = \begin{bmatrix} l_{11} & 0 & \cdots & 0 \\ l_{21} & l_{22} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ l_{n1} & \cdots & l_{n,n-1} & l_{nn} \end{bmatrix}, \mathbf{U} = \begin{bmatrix} \color{red}{1} & u_{12} & \cdots & u_{1n} \\ 0 & \color{red}{1} & \ddots & \vdots \\ \vdots & \ddots & \ddots & u_{n-1,n} \\ 0 & \cdots & 0 & \color{red}{1} \end{bmatrix}$$

Finding Elements of $\mathbf{L}$ and $\mathbf{U}$:

$$\mathbf{A} = \mathbf{LU} \quad \begin{bmatrix} 1 & u_{12} & u_{13} & \color{red}{u_{14}} \\ 0 & 1 & u_{23} & \color{red}{u_{24}} \\ 0 & 0 & 1 & \color{red}{u_{34}} \\ 0 & 0 & 0 & \color{red}{1} \end{bmatrix}$$

$$\begin{bmatrix} \color{red}{l_{11}} & \color{red}{0} & \color{red}{0} & \color{red}{0} \\ l_{21} & l_{22} & 0 & 0 \\ l_{31} & l_{32} & l_{33} & 0 \\ l_{41} & l_{42} & l_{43} & l_{44} \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} & \color{red}{a_{14}} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

$$l_{11} \cdot u_{14} + 0 \cdot u_{24} + 0 \cdot u_{34} + 0 \cdot 1 = a_{14}$$

$$\color{red}{u_{14}} = \color{red}{a_{14}} / \color{red}{l_{11}}$$

LU Decomposition Method

Crout Matrix Decomposition:

$$\mathbf{L} = \begin{bmatrix} l_{11} & 0 & \cdots & 0 \\ l_{21} & l_{22} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ l_{n1} & \cdots & l_{n,n-1} & l_{nn} \end{bmatrix}, \mathbf{U} = \begin{bmatrix} \mathbf{1} & u_{12} & \cdots & u_{1n} \\ 0 & \mathbf{1} & \ddots & \vdots \\ \vdots & \ddots & \ddots & u_{n-1,n} \\ 0 & \cdots & 0 & \mathbf{1} \end{bmatrix}$$

Finding Elements of $\mathbf{L}$ and $\mathbf{U}$:

$$\mathbf{A} = \mathbf{LU}$$

$$\begin{bmatrix} l_{11} & 0 & 0 & 0 \\ \mathbf{l_{21}} & \mathbf{l_{22}} & \mathbf{0} & \mathbf{0} \\ l_{31} & l_{32} & l_{33} & 0 \\ l_{41} & l_{42} & l_{43} & l_{44} \end{bmatrix} \begin{bmatrix} 1 & \mathbf{u_{12}} & u_{13} & u_{14} \\ 0 & \mathbf{1} & u_{23} & u_{24} \\ 0 & \mathbf{0} & 1 & u_{34} \\ 0 & \mathbf{0} & 0 & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & \mathbf{a_{22}} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

$$l_{21} \cdot u_{12} + l_{22} \cdot 1 + 0 \cdot 0 + 0 \cdot 0 = a_{22}$$

$$\mathbf{l_{22}} = \mathbf{a_{22}} - \mathbf{l_{21}} \cdot \mathbf{u_{12}}$$

LU Decomposition Method

Crout Matrix Decomposition:

$$\mathbf{L} = \begin{bmatrix} l_{11} & 0 & \cdots & 0 \\ l_{21} & l_{22} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ l_{n1} & \cdots & l_{n,n-1} & l_{nn} \end{bmatrix}, \mathbf{U} = \begin{bmatrix} \mathbf{1} & u_{12} & \cdots & u_{1n} \\ 0 & \mathbf{1} & \ddots & \vdots \\ \vdots & \ddots & \ddots & u_{n-1,n} \\ 0 & \cdots & 0 & \mathbf{1} \end{bmatrix}$$

Finding Elements of $\mathbf{L}$ and $\mathbf{U}$:

$$\mathbf{A} = \mathbf{LU}$$

$$\begin{bmatrix} l_{11} & 0 & 0 & 0 \\ l_{21} & l_{22} & 0 & 0 \\ \mathbf{l_{31}} & \mathbf{l_{32}} & \mathbf{l_{33}} & \mathbf{0} \\ l_{41} & l_{42} & l_{43} & l_{44} \end{bmatrix} \begin{bmatrix} 1 & \mathbf{u_{12}} & u_{13} & u_{14} \\ 0 & \mathbf{1} & u_{23} & u_{24} \\ 0 & \mathbf{0} & 1 & u_{34} \\ 0 & \mathbf{0} & 0 & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & \mathbf{a_{32}} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

$$l_{31} \cdot u_{12} + l_{32} \cdot 1 + l_{33} \cdot 0 + 0 \cdot 0 = a_{32}$$

$$\mathbf{l_{32}} = \mathbf{a_{32}} - \mathbf{l_{31}} \cdot \mathbf{u_{12}}$$

LU Decomposition Method

Crout Matrix Decomposition:

$$\mathbf{L} = \begin{bmatrix} l_{11} & 0 & \cdots & 0 \\ l_{21} & l_{22} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ l_{n1} & \cdots & l_{n,n-1} & l_{nn} \end{bmatrix}, \mathbf{U} = \begin{bmatrix} \mathbf{1} & u_{12} & \cdots & u_{1n} \\ 0 & \mathbf{1} & \ddots & \vdots \\ \vdots & \ddots & \ddots & u_{n-1,n} \\ 0 & \cdots & 0 & \mathbf{1} \end{bmatrix}$$

Finding Elements of $\mathbf{L}$ and $\mathbf{U}$:

$$\mathbf{A} = \mathbf{LU}$$

$$\begin{bmatrix} l_{11} & 0 & 0 & 0 \\ l_{21} & l_{22} & 0 & 0 \\ l_{31} & l_{32} & l_{33} & 0 \\ \mathbf{l_{41}} & \mathbf{l_{42}} & \mathbf{l_{43}} & \mathbf{l_{44}} \end{bmatrix} \begin{bmatrix} 1 & \mathbf{u_{12}} & u_{13} & u_{14} \\ 0 & \mathbf{1} & u_{23} & u_{24} \\ 0 & \mathbf{0} & 1 & u_{34} \\ 0 & \mathbf{0} & 0 & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & \mathbf{a_{42}} & a_{43} & a_{44} \end{bmatrix}$$

$$l_{41} \cdot u_{12} + l_{42} \cdot 1 + l_{43} \cdot 0 + l_{44} \cdot 0 = a_{42}$$

$$\mathbf{l_{42}} = \mathbf{a_{42}} - \mathbf{l_{41}} \cdot \mathbf{u_{12}}$$

LU Decomposition Method

Crout Matrix Decomposition:

$$\mathbf{L} = \begin{bmatrix} l_{11} & 0 & \cdots & 0 \\ l_{21} & l_{22} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ l_{n1} & \cdots & l_{n,n-1} & l_{nn} \end{bmatrix}, \mathbf{U} = \begin{bmatrix} \mathbf{1} & u_{12} & \cdots & u_{1n} \\ 0 & \mathbf{1} & \ddots & \vdots \\ \vdots & \ddots & \ddots & u_{n-1,n} \\ 0 & \cdots & 0 & \mathbf{1} \end{bmatrix}$$

Finding Elements of $\mathbf{L}$ and $\mathbf{U}$:

$$\mathbf{A} = \mathbf{LU}$$

$$\begin{bmatrix} l_{11} & 0 & 0 & 0 \\ \mathbf{l_{21}} & \mathbf{l_{22}} & \mathbf{0} & \mathbf{0} \\ l_{31} & l_{32} & l_{33} & 0 \\ l_{41} & l_{42} & l_{43} & l_{44} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & \mathbf{u_{13}} & u_{14} \\ 0 & 1 & \mathbf{u_{23}} & u_{24} \\ 0 & 0 & \mathbf{1} & u_{34} \\ 0 & 0 & \mathbf{0} & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & \mathbf{a_{23}} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

$$l_{21} \cdot u_{13} + l_{22} \cdot u_{23} + 0 \cdot 1 + 0 \cdot 0 = a_{23}$$

$$\mathbf{u_{23}} = (\mathbf{a_{23}} - \mathbf{l_{21}} \cdot \mathbf{u_{13}}) / \mathbf{l_{22}}$$

LU Decomposition Method

Crout Matrix Decomposition:

$$\mathbf{L} = \begin{bmatrix} l_{11} & 0 & \cdots & 0 \\ l_{21} & l_{22} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ l_{n1} & \cdots & l_{n,n-1} & l_{nn} \end{bmatrix}, \mathbf{U} = \begin{bmatrix} \color{red}{1} & u_{12} & \cdots & u_{1n} \\ 0 & \color{red}{1} & \ddots & \vdots \\ \vdots & \ddots & \ddots & u_{n-1,n} \\ 0 & \cdots & 0 & \color{red}{1} \end{bmatrix}$$

Finding Elements of $\mathbf{L}$ and $\mathbf{U}$:

$$\mathbf{A} = \mathbf{LU}$$

$$\begin{bmatrix} l_{11} & 0 & 0 & 0 \\ \color{red}{l_{21}} & \color{red}{l_{22}} & \color{red}{0} & \color{red}{0} \\ l_{31} & l_{32} & l_{33} & 0 \\ l_{41} & l_{42} & l_{43} & l_{44} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} & \color{red}{u_{14}} \\ 0 & 1 & u_{23} & \color{red}{u_{24}} \\ 0 & 0 & 1 & \color{red}{u_{34}} \\ 0 & 0 & 0 & \color{red}{1} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & \color{red}{a_{24}} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

$$l_{21} \cdot u_{14} + l_{22} \cdot u_{24} + 0 \cdot u_{34} + 0 \cdot 1 = a_{24}$$

$$\color{red}{u_{24}} = (a_{24} - l_{21} \cdot u_{14}) / l_{22}$$

LU Decomposition Method

Crout Matrix Decomposition:

$$\mathbf{L} = \begin{bmatrix} l_{11} & 0 & \cdots & 0 \\ l_{21} & l_{22} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ l_{n1} & \cdots & l_{n,n-1} & l_{nn} \end{bmatrix}, \mathbf{U} = \begin{bmatrix} \mathbf{1} & u_{12} & \cdots & u_{1n} \\ 0 & \mathbf{1} & \ddots & \vdots \\ \vdots & \ddots & \ddots & u_{n-1,n} \\ 0 & \cdots & 0 & \mathbf{1} \end{bmatrix}$$

Finding Elements of $\mathbf{L}$ and $\mathbf{U}$:

$$\mathbf{A} = \mathbf{LU}$$

$$\begin{bmatrix} l_{11} & 0 & 0 & 0 \\ l_{21} & l_{22} & 0 & 0 \\ \mathbf{l_{31}} & \mathbf{l_{32}} & \mathbf{l_{33}} & \mathbf{0} \\ l_{41} & l_{42} & l_{43} & l_{44} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & \mathbf{u_{13}} & u_{14} \\ 0 & 1 & \mathbf{u_{23}} & u_{24} \\ 0 & 0 & \mathbf{1} & u_{34} \\ 0 & 0 & \mathbf{0} & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & \mathbf{a_{33}} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

$$l_{31} \cdot u_{13} + l_{32} \cdot u_{23} + l_{33} \cdot 1 + 0 \cdot 0 = a_{33}$$

$$\mathbf{l_{33}} = \mathbf{a_{33}} - \mathbf{l_{31}} \cdot \mathbf{u_{13}} - \mathbf{l_{32}} \cdot \mathbf{u_{23}}$$

LU Decomposition Method

Crout Matrix Decomposition:

$$\mathbf{L} = \begin{bmatrix} l_{11} & 0 & \cdots & 0 \\ l_{21} & l_{22} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ l_{n1} & \cdots & l_{n,n-1} & l_{nn} \end{bmatrix}, \mathbf{U} = \begin{bmatrix} \mathbf{1} & u_{12} & \cdots & u_{1n} \\ 0 & \mathbf{1} & \ddots & \vdots \\ \vdots & \ddots & \ddots & u_{n-1,n} \\ 0 & \cdots & 0 & \mathbf{1} \end{bmatrix}$$

Finding Elements of $\mathbf{L}$ and $\mathbf{U}$:

$$\mathbf{A} = \mathbf{LU}$$

$$\begin{bmatrix} l_{11} & 0 & 0 & 0 \\ l_{21} & l_{22} & 0 & 0 \\ l_{31} & l_{32} & l_{33} & 0 \\ \mathbf{l_{41}} & \mathbf{l_{42}} & \mathbf{l_{43}} & \mathbf{l_{44}} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & \mathbf{u_{13}} & u_{14} \\ 0 & 1 & \mathbf{u_{23}} & u_{24} \\ 0 & 0 & \mathbf{1} & u_{34} \\ 0 & 0 & \mathbf{0} & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & \mathbf{a_{43}} & a_{44} \end{bmatrix}$$

$$l_{41} \cdot u_{13} + l_{42} \cdot u_{23} + l_{43} \cdot 1 + l_{44} \cdot 0 = a_{43}$$

$$\mathbf{l_{43}} = \mathbf{a_{43}} - \mathbf{l_{41}} \cdot \mathbf{u_{13}} - \mathbf{l_{42}} \cdot \mathbf{u_{23}}$$

LU Decomposition Method

Crout Matrix Decomposition:

$$\mathbf{L} = \begin{bmatrix} l_{11} & 0 & \cdots & 0 \\ l_{21} & l_{22} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ l_{n1} & \cdots & l_{n,n-1} & l_{nn} \end{bmatrix}, \mathbf{U} = \begin{bmatrix} \mathbf{1} & u_{12} & \cdots & u_{1n} \\ 0 & \mathbf{1} & \ddots & \vdots \\ \vdots & \ddots & \ddots & u_{n-1,n} \\ 0 & \cdots & 0 & \mathbf{1} \end{bmatrix}$$

Finding Elements of $\mathbf{L}$ and $\mathbf{U}$:

$$\mathbf{A} = \mathbf{LU}$$

$$\begin{bmatrix} l_{11} & 0 & 0 & 0 \\ l_{21} & l_{22} & 0 & 0 \\ \mathbf{l_{31}} & \mathbf{l_{32}} & \mathbf{l_{33}} & \mathbf{0} \\ l_{41} & l_{42} & l_{43} & l_{44} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} & \mathbf{u_{14}} \\ 0 & 1 & u_{23} & \mathbf{u_{24}} \\ 0 & 0 & 1 & \mathbf{u_{34}} \\ 0 & 0 & 0 & \mathbf{1} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & \mathbf{a_{34}} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

$$l_{31} \cdot u_{14} + l_{32} \cdot u_{24} + l_{33} \cdot u_{34} + 0 \cdot 1 = a_{34}$$

$$\mathbf{u_{34}} = (a_{34} - l_{31} \cdot u_{14} - l_{32} \cdot u_{24}) / l_{33}$$

LU Decomposition Method

Crout Matrix Decomposition:

$$\mathbf{L} = \begin{bmatrix} l_{11} & 0 & \cdots & 0 \\ l_{21} & l_{22} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ l_{n1} & \cdots & l_{n,n-1} & l_{nn} \end{bmatrix}, \mathbf{U} = \begin{bmatrix} \mathbf{1} & u_{12} & \cdots & u_{1n} \\ 0 & \mathbf{1} & \ddots & \vdots \\ \vdots & \ddots & \ddots & u_{n-1,n} \\ 0 & \cdots & 0 & \mathbf{1} \end{bmatrix}$$

Finding Elements of $\mathbf{L}$ and $\mathbf{U}$:

$$\mathbf{A} = \mathbf{LU}$$

$$\begin{bmatrix} l_{11} & 0 & 0 & 0 \\ l_{21} & l_{22} & 0 & 0 \\ l_{31} & l_{32} & l_{33} & 0 \\ \mathbf{l_{41}} & \mathbf{l_{42}} & \mathbf{l_{43}} & \mathbf{l_{44}} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} & \mathbf{u_{14}} \\ 0 & 1 & u_{23} & \mathbf{u_{24}} \\ 0 & 0 & 1 & \mathbf{u_{34}} \\ 0 & 0 & 0 & \mathbf{1} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & \mathbf{a_{44}} \end{bmatrix}$$

$$l_{41} \cdot u_{14} + l_{42} \cdot u_{24} + l_{43} \cdot u_{34} + l_{44} \cdot 1 = a_{44}$$

$$\mathbf{l_{44}} = \mathbf{a_{44}} - \mathbf{l_{41}} \cdot \mathbf{u_{14}} - \mathbf{l_{42}} \cdot \mathbf{u_{24}} - \mathbf{l_{43}} \cdot \mathbf{u_{34}}$$

LU Decomposition Method

Example 03-04: Find root of the following system of equations.

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 3 \\ 6 \\ 5 \end{Bmatrix}$$

LU Decomposition Method

Example 03-04: Find root of the following system of equations.

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 3 \\ 6 \\ 5 \end{Bmatrix}$$

Crout Matrix Decomposition:

LU Decomposition Method

Example 03-04: Find root of the following system of equations.

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 3 \\ 6 \\ 5 \end{Bmatrix}$$

Crout Matrix Decomposition:

$$\mathbf{A} = \mathbf{LU} \quad \rightarrow \quad \begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

LU Decomposition Method

Example 03-04: Find root of the following system of equations.

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 3 \\ 6 \\ 5 \end{Bmatrix}$$

Crout Matrix Decomposition:

$$\mathbf{A} = \mathbf{LU} \quad \rightarrow \quad \begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{bmatrix}$$

LU Decomposition Method

Example 03-04: Find root of the following system of equations.

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 3 \\ 6 \\ 5 \end{Bmatrix}$$

Crout Matrix Decomposition:

$$\mathbf{A} = \mathbf{LU} \quad \rightarrow \quad \begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{bmatrix} \quad \begin{aligned} l_{11} \cdot 1 + 0 \cdot 0 + 0 \cdot 0 &= 1 \\ l_{11} &= 1 \end{aligned}$$

LU Decomposition Method

Example 03-04: Find root of the following system of equations.

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 3 \\ 6 \\ 5 \end{Bmatrix}$$

Crout Matrix Decomposition:

$$\mathbf{A} = \mathbf{LU} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{bmatrix} \quad \begin{aligned} l_{21} \cdot 1 + l_{22} \cdot 0 + 0 \cdot 0 &= 2 \\ l_{21} &= 2 \end{aligned}$$

LU Decomposition Method

Example 03-04: Find root of the following system of equations.

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 3 \\ 6 \\ 5 \end{Bmatrix}$$

Crout Matrix Decomposition:

$$\mathbf{A} = \mathbf{LU} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{bmatrix} \quad \begin{aligned} l_{31} \cdot 1 + l_{32} \cdot 0 + l_{33} \cdot 0 &= 3 \\ l_{31} &= 3 \end{aligned}$$

LU Decomposition Method

Example 03-04: Find root of the following system of equations.

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 3 \\ 6 \\ 5 \end{Bmatrix}$$

Crout Matrix Decomposition:

$$\mathbf{A} = \mathbf{LU} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{1} & \mathbf{0} & \mathbf{0} \\ 2 & l_{22} & 0 \\ 3 & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} 1 & \mathbf{u_{12}} & u_{13} \\ 0 & \mathbf{1} & u_{23} \\ 0 & \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} 1 & \mathbf{2} & 1 \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{bmatrix}$$

$$1 \cdot u_{12} + 0 \cdot 1 + 0 \cdot 0 = 2$$
$$\mathbf{u_{12} = 2/1 = 2}$$

LU Decomposition Method

Example 03-04: Find root of the following system of equations.

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 3 \\ 6 \\ 5 \end{Bmatrix}$$

Crout Matrix Decomposition:

$$\mathbf{A} = \mathbf{LU} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{1} & \mathbf{0} & \mathbf{0} \\ 2 & l_{22} & 0 \\ 3 & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & \mathbf{u_{13}} \\ 0 & 1 & \mathbf{u_{23}} \\ 0 & 0 & \mathbf{1} \end{bmatrix} \begin{bmatrix} 1 & 2 & \mathbf{1} \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{bmatrix}$$

$$1 \cdot u_{13} + 0 \cdot u_{23} + 0 \cdot 1 = 1$$
$$\mathbf{u_{13}} = \mathbf{1/1 = 1}$$

LU Decomposition Method

Example 03-04: Find root of the following system of equations.

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 3 \\ 6 \\ 5 \end{Bmatrix}$$

Crout Matrix Decomposition:

$$\mathbf{A} = \mathbf{LU} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & l_{22} & 0 \\ 3 & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{aligned} 2 \cdot 2 + l_{22} \cdot 1 + 0 \cdot 0 &= -1 \\ l_{22} &= -1 - 2 \cdot 2 = -5 \end{aligned}$$

LU Decomposition Method

Example 03-04: Find root of the following system of equations.

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 3 \\ 6 \\ 5 \end{Bmatrix}$$

Crout Matrix Decomposition:

$$\mathbf{A} = \mathbf{LU} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & l_{22} & 0 \\ 3 & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

$$3 \cdot 2 + l_{32} \cdot 1 + l_{33} \cdot 0 = 1$$
$$l_{32} = 1 - 3 \cdot 2 = -5$$

LU Decomposition Method

Example 03-04: Find root of the following system of equations.

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 3 \\ 6 \\ 5 \end{Bmatrix}$$

Crout Matrix Decomposition:

$$\mathbf{A} = \mathbf{LU} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & -5 & 0 \\ 3 & -5 & l_{33} \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

$$2 \cdot 1 + (-5) \cdot u_{23} + 0 \cdot 1 = 2$$
$$u_{23} = (2 - 2 \cdot 1) / (-5) = 0$$

LU Decomposition Method

Example 03-04: Find root of the following system of equations.

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 3 \\ 6 \\ 5 \end{Bmatrix}$$

Crout Matrix Decomposition:

$$\mathbf{A} = \mathbf{LU} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & -5 & 0 \\ 3 & -5 & l_{33} \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{aligned} 3 \cdot 1 + (-5) \cdot 0 + l_{33} \cdot 1 &= -1 \\ l_{33} &= -1 - 3 \cdot 1 - (-5) \cdot 0 = -4 \end{aligned}$$

LU Decomposition Method

Example 03-04: Find root of the following system of equations.

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 3 \\ 6 \\ 5 \end{Bmatrix}$$

Crout Matrix Decomposition:

$$\mathbf{A} = \mathbf{LU} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & -5 & 0 \\ 3 & -5 & -4 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Root of System of Equations:

$$\mathbf{Ax} = \mathbf{LUx} = \mathbf{b} \rightarrow \mathbf{Ly} = \mathbf{b} \rightarrow \mathbf{Ux} = \mathbf{y}$$

$\mathbf{Ly} = \mathbf{b}$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & -5 & 0 \\ 3 & -5 & -4 \end{bmatrix} \begin{Bmatrix} y_1 \\ y_2 \\ y_3 \end{Bmatrix} = \begin{Bmatrix} 3 \\ 6 \\ 5 \end{Bmatrix}$$
$$\begin{aligned} y_1 &= \frac{3}{1} = 3 \\ y_2 &= \frac{6 - 2 \cdot y_1}{-5} = \frac{6 - 2 \cdot 3}{-5} = 0 \\ y_3 &= \frac{5 - 3 \cdot y_1 - (-5) \cdot y_2}{-4} = \frac{5 - 3 \cdot 3 - (-5) \cdot 0}{-4} = 1 \end{aligned}$$

$\mathbf{Ux} = \mathbf{y}$

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 3 \\ 0 \\ 1 \end{Bmatrix}$$
$$\begin{aligned} x_3 &= 1 \\ x_2 &= 0 - 0 \cdot x_3 = 0 \\ x_1 &= 3 - 2 \cdot x_2 - 1 \cdot x_3 = 2 \end{aligned}$$